

System overview

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3.1 What the unit does

The SalinBloc™ unit is built around HydroVolta's SonixED™ technology, which combines three mechanisms in one automated platform:

1. **Electrodialysis reversal (EDR):** an applied DC electric field draws dissolved ions — not water — through ion-exchange membranes from the diluate stream into the concentrate stream. Because water itself does not have to be forced through a membrane (unlike reverse osmosis), the process runs at much lower hydraulic pressure. Periodically reversing polarity cleans the membranes in place without chemicals and gives more stable performance as feed salinity varies.
2. **Integrated ultrasound antifouling:** ultrasound transducers built into the stack create acoustic cavitation at the membrane surface, disrupting scale nucleation and helping prevent silica fouling — this is the element of SonixED™ that HydroVolta holds a granted European patent on (2021).
3. **Custom membrane coating:** a proprietary coating gives the membranes monovalent-ion selectivity, so the process can target nitrate, sodium, chloride and fluoride removal while retaining calcium and magnesium in the product water.

The result is a saline feed stream split into a low-salinity product stream (*diluate*) and a concentrated reject stream (*concentrate/brine*), with chemical-free ultrasound antifouling keeping the membrane stack clean and extending electrode/membrane life (chapter 10). Typical applications: groundwater desalination and selective ion (e.g. nitrate) removal, industrial process-water reuse, brine valorization/concentration, and Zero Liquid Discharge (ZLD) support.



Figure 3.1. Exterior view of a delivered SalinBloc™ container on site.

3.2 Field performance reference

HydroVolta publishes the following performance comparison from a 90-day industrial pilot (February–May 2023, cooling-tower blowdown at a chemical industry facility in Germany, feed TDS $\approx$ 4,500 ppm, silica 85 mg/l, high hardness), run head-to-head against reverse osmosis (RO) and a standard EDR system on the same feedwater (source: hydrovolta.com/technology.html and [/projects.html](https://hydrovolta.com/projects.html)). These figures describe that specific pilot and HydroVolta's general SonixED™ technology, not a performance guarantee for any individual SalinBloc™ unit; use them as context for what the technology is designed to achieve, and always verify against the site-specific commissioning data recorded in appendix A.

Table 3.1. Germany 2023 industrial pilot — SonixED™ vs. RO vs. standard EDR (source: hydrovolta.com)

Parameter	SonixED™	RO	Standard EDR
Water recovery	90–98%	50–75%	70–85%
Energy consumption	0.19– 0.7 kWh/m ³	2.0– 4.0 kWh/m ³	1.0– 2.0 kWh/m ³
Operating pressure	0.5–2.5 bar	25–35 bar	1–2 bar
Chemical dosing	Minimal, no antiscalant	Required	Antiscalant needed
CIP frequency	0.6×/month	1×/month	1×/month
Total treatment cost	\$0.30– 0.40/m ³	\$0.60– 1.00/m ³	\$0.40–0.80/m ³

HydroVolta states SonixED™'s general operating envelope as feed TDS 1,500–20,000 ppm (“strong fit” also for high silica >30 mg/l and high hardness >300 mg/l Ca+Mg feeds); seawater (>35,000 ppm) is stated as not the primary use case. Active and piloted SonixED™/SalinBloc™ deployments referenced by HydroVolta include a flagship groundwater desalination project in Morocco (seawater-intruded coastal aquifer, $\approx$ 7,000 ppm feed), a multi-site utility pilot with Water Corporation in Perth, Australia (selective nitrate and salinity removal), the Germany industrial pilot above, and earlier-stage projects in Brazil and Saudi Arabia.

3.3 Container layout

The **standard SalinBloc™ unit is a single container** (20 ft or 40 ft, High-Cube or Standard) housing the SonixED™ skid(s) (chapters 9 to 11 and 14) and everything downstream of the feed connection — it does **not** include a pretreatment stage, so feed water must already meet the stack's feedwater envelope (chapter 10) before it reaches the unit. **Pretreatment (chapter 8) is an optional add-on**, not part of the standard product: where a site's raw water needs filtration first, HydroVolta can supply a *second, separate pretreatment container*, connected hydraulically ahead of the standard unit. Everything below describes the standard single-container unit unless a step is explicitly marked as part of the optional pretreatment container. A documented 20 ft general-arrangement layout (drawing 2308CA1) shows internal dimensions of 6058 × 2438 × 2896 mm, housing a bank of SonixED™ stack racks, an electrical/control cabinet, a dosing/chemical section, three overhead storage tanks, and interconnecting piping; a companion top-view drawing (2308CA2) lays the skids out along the container floor: chemical skid, pump skid, tank skid, control 1 skid, filter skid, SonixED™ skid, control 2 skid.

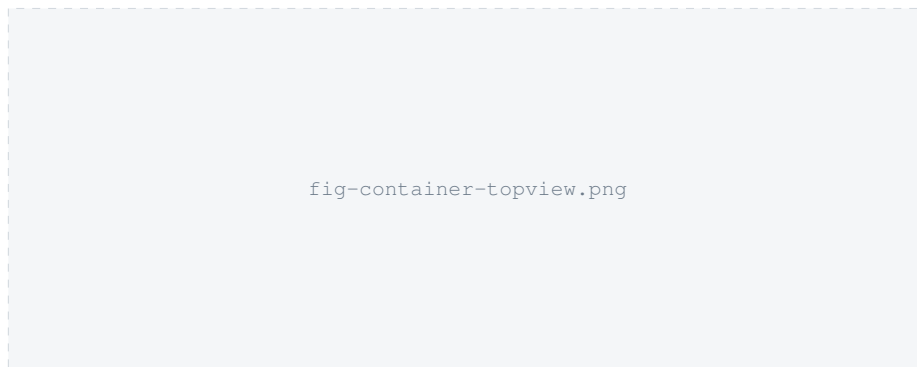


Figure 3.2. Suggested figure: the skid top-view layout (based on drawing 2308CA2), redrawn/simplified for this manual, with each skid numbered and cross-referenced to its Part II chapter.

3.4 Process flow

Standard unit (no pretreatment):

1. Feed water enters the container directly (2" connection) — it must already meet the stack's feedwater envelope (chapter 10); the unit does not filter or otherwise condition it first.
2. Feed water is split and pumped through parallel SonixED™ stacks (chapters 9 and 10), with three internal streams per stack: **diluate** (becomes the product/permeate), **concentrate** (becomes the brine/reject), and **electrolyte** (a separate loop feeding the electrode compartments, chapter 18).
3. Product (permeate) water exits the container via a 2" outlet connection.
4. Reject/concentrate is routed to a drain tank via a 3" drainage connection.

With the optional pretreatment container added (only relevant if that add-on was ordered, chapter 8): feed water instead enters the separate pretreatment container first (2½" connection from source); disc filtration and ultrafiltration condition it there; filtered water then leaves via a 2" connection into the main unit's feed inlet above, and the pretreatment container's own backwash/ drainage is tee'd together with the main unit's drain line to the shared drain tank.

3.5 Main components

This section is a one-page index of every compartment covered in its own chapter later in this manual (Part II) — use it to jump straight to the part you need.

- **Pretreatment and filtration skid (optional add-on, not part of the standard unit)** — chapter 8
- **Feed and process pumps** — chapter 9
- **SonixED™ stack modules** — chapter 10
- **Valves and manifold** — chapter 11
- **Dosing system** — chapter 12
- **Instrumentation and sensors** — chapter 13
- **Control cabinet, PLC and HMI** — chapter 14
- **Diluate tank** — chapter 15
- **Concentrate tank** — chapter 16
- **CIP tank** — chapter 17
- **Electrolyte tank** — chapter 18
- **Telemetry and remote connectivity** — chapter 19

3.6 Control cabinet and HMI

The control system consists of a PLC with a touchscreen HMI, manual/override controls, and RS485 communication, housed in an IP55-rated cabinet. The skid ordering code allows two control-box configurations — **CB0** (PLC only) and **CB1** (PLC + touchscreen HMI) — and three cloud-connectivity tiers: **CL0** (local only), **CL1** (cloud monitoring), **CL2** (full cloud monitoring and control). See chapter 14 for details.

3.7 Nameplate and identification

TODO: no nameplate drawing, CE-marking template with filled-in fields, or Declaration of Conformity content was found (the 02_DoC's and 05_SonixED_stack CE-marking folders in HydroVolta's engineering library are confirmed empty — see appendix C). The confirmed identification data is the datasheet labeling convention "Model: SONIXED-SKID, Version: 2025-A7" and the ordering code format SONIXED-SKID-A-S-M-PR-V-CB-CL (application type A0/A1/A2, stack count 1/2/3, measuring points 6/9/12, pressure sensors 6/9/12, valve count 2/4/6, control-box option CB0/CB1, cloud option CL0/CL1/CL2). Each skid physically measures approximately $3040 \times 1340 \times 2020$ mm (l × b × h), on an aluminum-profile frame with a PVC/thermoplastic-spacer housing, mounted on lockable wheels for in-container mobility. HydroVolta's public SalinBloc™ brochure states container throughput by format: a **20 ft** unit handles 2–30 m³/h (≈ 44 –660 m³/day at 22 h/day operation); a **40 ft** unit handles 15–60 m³/h (≈ 330 –1,320 m³/day) — see section 3.2 and appendix A. No container weight figure was found in any source reviewed.



Figure 3.3. Suggested figure: photograph of the unit's nameplate once affixed, showing serial number, manufacture date, voltage/power rating, IP rating and CE mark.